Amutheezan Sivagnanam

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SUMMARY

PhD Candidate and **Machine Learning Researcher** specializing in **Machine Learning** with **six** years of academic research and **two** years of industrial software development experience. Expertise in **scalable deep learning and reinforcement learning systems**, ML **training, and deployment**, complemented by a **strong publication record (h-index 6)**. Proficient in **Python**, **TensorFlow**, **PyTorch**, AWS, and agile CI/CD practices, with hands-on experience in **optimizing ML models and pipelines**.

EDUCATION

Pennsylvania State University
Ph.D., Computer Science — *Aug 2025*

University of Houston
M.S., Computer Science — *Aug 2022*

University of Moratuwa
B.S., (Hons) Engineering (Computer Science and Engineering) — *Jan 2018*

AREAS OF EXPERTISE

Reinforcement Learning, Optimization, Operational Research

RESEARCH EXPERIENCE

Pennsylvania State University — Graduate Research Assistant
Applied Artificial Intelligence Lab
Aug. 2022 - May 2025

- Spearheaded research projects funded by the U.S. Department of Energy (DOE) and the National Science Foundation (NSF).

- Studied real-world decision-making problems and identified gaps in existing solution approaches.
- Developed mathematical models and formulated problems to address these challenges.
- Applied artificial intelligence–based solutions to tackle real-world problems and successfully deployed them in relevant industries.
- Published research findings in top-tier AI conferences (ICML) and assisted preparing slides for presenting results at DOE and NSF meetings.
- Developed a deep reinforcement learning (DRL) method for proactive responder repositioning in emergency management, achieving response times 1000× faster while reducing operational delays.
- Applied DRL to solve online vehicle routing with advance booking, enabling real-time confirmations within seconds.

The University of Houston — Graduate Research Assistant
Resilient Networks and Systems Lab
Sep. 2019 - Aug. 2022

- Spearheaded research projects funded by the U.S. Department of Energy (DOE) and the National Science Foundation (NSF).
- Studied real-world decision-making problems and identified gaps in existing solution approaches.
- Developed mathematical models and formulated problem statements to effectively address these challenges.
- Applied AI-based solution approaches to real-world problems and successfully deployed them in relevant industries.
- Published research findings in top-tier AI conferences (AAAI, IJCAI) and assisted preparing slides for presenting results at DOE and NSF meetings.
- Introduced heuristics that reduced annual energy costs by $140K for public transit agencies operating mixed fleets of buses.
- Implemented a deep reinforcement learning approach that reduced operational costs by 20% by enabling online booking for traditionally offline Vehicle Routing Problems (VRPs).
- Collected and analyzed real-world data using Python to identify trends—such as changes in paratransit operations before and after COVID-19 and the impact of vulnerability reward programs.

SOFTWARE ENGINEERING EXPERIENCE

LSEG Technology — Software Engineer
Post Trade
Jan 2018 – Jul 2019

- Writing application software in the Object-Oriented manner.
- Practicing agile development practices.
- Working experience with languages such as Java, Python, and C++
- Introduce Unit Testing for Libraries in Post Trade C++ Code
- Worked on Making changes in DB and Tested with BDD based Testing.
- Worked on Duplicated Error Codes Identification and Replacement with valid New Error Codes.
- Developed DSO for Front-End End to End Testing.
- Analyzed and Changed Back-End Regression Script to work with both Old and New Testing Framework
- Developed a New Report Generation for Testing Framework, which used to be emailed at the end of regression.
- Completed the Code Integration works related to Back-End Regression
- Updated automatic updates to auto-generated codes based on Database changes using Integration Plans.
- Participated in Code Integration and Code Deployment Plans.
- Participated in professional training programs conducted by Millennium IT Software and Post Trade Team.
- Worked on Front-End Development for both Product and Solution which consists of Enhancement, Bug Fixing, Merging and Introducing new features.

WSO2 Lanka PVT Ltd — Software Engineering Intern
Data Analytics Team
Jul 2016 – Dec 2016

- Developed an alert generation system analyzing descriptive HL7/FHIR data to monitor disease outbreaks, triggering timely email and SMS notifications.
- Engineered a mechanism to evaluate hospital functionality by assessing admission and discharge messages, including bed and oxygen cylinder availability.

SKILLS

Programming Languages: Python, C++, Java, C, JavaScript, PHP, MATLAB
Frameworks & Libraries: PyTorch, TensorFlow, Scikit-learn, Transformers, Deep Learning, Reinforcement Learning, DQN, DDPG, TRPO, PPO, GRPO, RLHF, Bayesian Optimization, Clustering, Statistical Analysis, Data Mining, Ray, RLLib, OpenAI Gym, NumPy, pandas, matplotlib
Databases: SQL, MySQL, SQLite, MongoDB, Oracle DB
Tools & Platforms: Amazon Web Services, Git, GitHub, Linux, Docker, Spark, CI/CD, Bash
Methodologies & Practices: Object-Oriented Development, Agile Practices, Behaviour Driven Testing
Optimization & Operations: Linear Programming, Integer Programming, CPLEX, Gurobi, Mosek, OR-Tools

PUBLICATIONS

CONFERENCE PROCEEDINGS

[C7] **Sivagnanam, A.,** Pettet, A., Lee, H., Mukhopadhyay, A., Dubey, A., & Laszka, A. (2024). Multi-agent reinforcement learning with hierarchical coordination for emergency responder stationing. In *Proceedings of the International Conference on Machine Learning (ICML-24)*.

[C6] R. Sen, **A. Sivagnanam,** A. Laszka, A. Mukhopadhyay and A. Dubey, "Grid-Aware Charging and Operational Optimization for Mixed-Fleet Public Transit," *2024 IEEE 27th International Conference on Intelligent Transportation Systems (ITSC)*, Edmonton, AB, Canada, 2024, pp. 4172-4179,

[C5] Pavia, S., Rogers, D., **Sivagnanam, A**., Wilbur, M., Edirimanna, D., Kim, Y., Mukhopadhyay, A., Pugliese, P., Samaranayake, S., Laszka, A., & Dubey, A. (2024). SmartTransit.AI: A dynamic paratransit and microtransit application. In *Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence (IJCAI-24)* (pp. 8767–8770).

[C4] Atefi, S., **Sivagnanam, A.,** Ayman, A., Grossklags, J., & Laszka, A. (2023, May). The benefits of vulnerability discovery and bug bounty programs: Case studies of Chromium and Firefox. In *Proceedings of the ACM Web Conference* (pp. 2209–2219).

[C3] **Sivagnanam, A**., Kadir, S. U., Mukhopadhyay, A., Pugliese, P., Dubey, A., Samaranayake, S., & Laszka, A. (2022, July). Offline vehicle routing problem with online bookings: A novel problem formulation with applications to paratransit. In *Proceedings of the Thirty-First International Joint Conference on Artificial Intelligence (IJCAI-22)* (pp. 3933–3939).

[C2] **Sivagnanam, A**., Ayman, A., Wilbur, M., Pugliese, P., Dubey, A., & Laszka, A. (2021, May). Minimizing energy use of mixed-fleet public transit for fixed-route service. In *Proceedings of the AAAI Conference on Artificial Intelligence (Vol. 35, No. 17, pp. 14930–14938).*

[C1] Ayman, A., Wilbur, M., **Sivagnanam, A.,** Pugliese, P., Dubey, A., & Laszka, A. (2020, September). *Data-driven prediction of route-level energy use for mixed-vehicle transit fleets.* In Proceedings of the 6th IEEE International Conference on Smart Computing (SMARTCOMP) (pp. 1–8). IEEE.

JOURNALS

[J2] Wilbur, M., Ayman, A., **Sivagnanam, A.,** Ouyang, A., Poon, V., Kabir, R., Vadali, A., Pugliese, P., Freudberg, D., Laszka, A., & Dubey, A. (2023). Impact of COVID-19 on Public Transit Accessibility and Ridership. *Transportation Research Record*, *2677*(4), 531-546.

[J1] Ayman, A., **Sivagnanam, A**., Wilbur, M., Pugliese, P., Dubey, A., & Laszka, A. (2022). Data-driven prediction and optimization of energy use for transit fleets of electric and ICE vehicles. *ACM Transactions on Internet Technology, 22*(1), Article 7, 1–29

WORKSHOPS

[W1] **Sivagnanam, A.,** Atefi, S., Ayman, A., Grossklags, J., & Laszka, A. (2021). On the benefits of bug bounty programs: A study of Chromium vulnerabilities. In *Workshop on the Economics of Information Security (WEIS)* (Vol. 10).

BOOK CHAPTERS

[BC1] Wilbur, M., **Sivagnanam, A.,** Ayman, A., Samaranayake, S., Dubey, A., & Laszka, A. (2023, August). Artificial intelligence for smart transportation. In **Y. Vorobeychik & A. Mukhopadhyay (Eds.),** *Artificial Intelligence and Society* (book chapter). ACM Press.

TALKS

IJCAI 2022 Vienna, Austria

For the paper: Offline Vehicle Routing Problem with Online Bookings: A Novel Problem Formulation with Applications to Paratransit

WEIS 2021 Virtual Workshop

For the workshop paper: On the Benefits of Bug Bounty Programs:
A Study of Chromium Vulnerabilities

AAAI 2021 Virtual Conference

For the paper: Minimizing Energy Use of Mixed-Fleet Public Transit
for Fixed-Route Service

SERVICES

Program Committee Member
International Joint Conference on Artificial Intelligence (AI For Social Good) 2025

Reviewer
International Joint Conference on Artificial Intelligence (AI For Social Good) 2024

Reviewer
AI4Research Workshop @IJCAI2024 2024

Auxiliary Reviewer
22nd International Conference on Autonomous Agents and Multiagent Systems 2023

AWARDS AND HONORS

Information Security Quiz 2015 - Runners Up 2015
Issued by Sri Lanka Computer Emergency Readiness Team(SLCERT)
and Information and Communication Technology Agency of Sri Lanka(ICTA)

Mahapola Scholarship 2014 - 2017